Methodology 1 — Boolean relaxation

From discrete subsets to the cardinality polytope

Table of contents

> *Draft outline — content to come.*

Reparameterising the support

Replace the discrete indicator $z \in \{0,1\}^p$ by a continuous $t \in [0,1]^p$ and write

$$\beta_j = t_j\,\xi_j, \qquad t_j \in [0,1].$$

- $t_j = 1$ means feature j is **in**.
- $t_j = 0$ means **out**.
- Vertices of the hypercube discrete supports.

The cardinality constraint $\sum z_j \le k$ becomes $\sum t_j \le k$ — a polytope.

The relaxed objective

$$f_{\delta,\lambda}(t) = \min_{\beta_0,\,\xi}\left\{-\frac{1}{n}\,\ell(\beta_0, T_t\xi) + \lambda\,\|\xi\|_2^2 + \delta\,\|\Gamma_t\,\xi\|_2^2\right\}.$$

- Inner problem in ξ: **ridge GLM**, solved by `glmnet` in one call.
- Outer problem in t: smooth, defined on the cardinality polytope.
- δ is a curvature parameter (homotopy schedule lives here).

Envelope gradient

By Danskin's theorem, the gradient w.r.t. t has closed form:

$$g_j(t) \;=\; -\frac{2(\lambda+\delta)}{t_j^3}\,\hat{\xi}_j(t)^2.$$

This is the only thing the outer Frank-Wolfe step needs.

> Placeholder for visual: the four-panel loss-landscape figure (discrete $\to$ smooth $\to$ smooth+ℓ_1 $\to$ strongly penalised).

← COMBSS preview · Next → Frank-Wolfe